

Student Admissions Portal - User Guide & Workflow Manual

This manual explains how the AI-Enabled Student Admission System works from a functional perspective. It covers the user journeys for both prospective students and counselor administrators, detailing how they interact with each module.

1. Prospective Student Journey

The student journey is designed to be self-explanatory, taking applicants from initial inquiry on the landing page to a successful admission registration ticket.

Step 1: Navigating the Landing Page

- **User Interaction:**
- The student visits the landing page (<http://localhost:3001/>).
- They can view placement success rates, facilities, and available programs.
- They can use the **Program Finder** dropdowns to filter available degrees and areas of study relative to the main Bhimavaram campus.
- Clicking **Enroll Now** or **Find Your Program** redirects them to the multi-step admission wizard.

Step 2: Form Intake (Steps 1 & 2)

- **User Interaction:**
- **Step 1 (Personal Info):** The candidate enters details like their Full Name, Date of Birth, Gender, Nationality (e.g. Indian), and Category (General, OBC, SC, ST).
- **Step 2 (Academic Profile):** The candidate inputs their 10th and 12th details. They must choose their **Entrance Exam Type** (AP EAPCET for first-year admission, or AP ECET for lateral diploma second-year entry) and input their respective rank score.

Step 3: Branch Selection & Merit Cutoff Validation (Step 3)

- **User Interaction:**
- The student selects their **Primary Branch Preference** (e.g., Computer Science & Engineering) and **Secondary Branch Preference**.
- **Dynamic Cutoff Checks:** The page automatically verifies their entrance rank against institutional cutoffs for their selected course:
- E.g., The AP EAPCET merit cutoff for CSE is **8,000**. If the candidate enters rank **12,000** and selects CSE as their primary choice under the Merit Quota, the form displays a red warning and blocks progress:

 "Your AP EAPCET Rank (12000) exceeds the Merit Quota cutoff for CSE (8000). Please select another course or apply under General/No Scholarship."

Step 4: Quota Selection & Live Fee Calculator

- **User Interaction:**
- The student chooses a **Scholarship Quota / Category** from a select dropdown:
- *General / No Scholarship*: flat 0% Discount.
- *Sports Quota*: flat 20% Discount.
- *EWS (Economically Weaker Section)*: flat 30% Discount.
- *Board Topper (12th Grade 95%+)*: flat 50% Discount.
- *Merit Quota*: Dynamically calculates a discount percentage based on their competitive entrance rank score (under 5k rank = 50% waiver, under 15k = 30%, under 30k = 15%).
- **Interactive Widget:** As they toggle quotas, the live fee invoice updates at the bottom showing:
 - **Standard Tuition Fee** (e.g., ₹120,000)
 - **Applied Discount** (e.g., 30%)
 - **Final Payable Fee** (e.g., ₹84,000)

Step 5: Submission & Ticket Receipt

- **User Interaction:**
 - The student selects the data authorization checklist and clicks **Submit**.
 - The application is written directly to the PostgreSQL database.
 - The student is presented with a success confirmation card with their unique student reference ticket (e.g. `VITB2026-A48293`).
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2. Counselor Administrator Journey

The counselor portal allows university admissions officers to securely review, inspect, and approve pending applications.

Step 1: Secure Administrator Login

- **User Interaction:**
- The counselor goes to `/counselor/login` .
- Enters secure administrative credentials.
- On submission, the backend issues an HMAC-SHA256 authenticated security web token (JWT) which is stored for that browser session (expiring in 1 hour).

Step 2: Dashboard Queue & Application Inspection

- **User Interaction:**
- The counselor is redirected to their Dashboard.
- They see a list of all applicants fetched dynamically from the database.
- **Search & Filter Actions:** The counselor can type names in the search bar, filter by course branch, or isolate candidates by their approval status (Approved, Pending, Rejected).
- Clicking on a candidate's line item opens the applicant details panel list.

Step 3: Triggering AI Candidate analysis

- **User Interaction:**

- The counselor clicks the **AI Insights** button on a student's card.
- **Behind the Scenes:** The system queries Google Gemini AI using applicant data. Gemini evaluates qualifying credentials, discount parameters, and flags potential risks (such as a high discount requested with marginal entrance marks).
- The AI returns status recommendations (e.g., "Normal priority" or "Requires Manager Override due to 40% discount").

Step 4: Verification Override Actions

- **User Interaction:**
- The counselor decides to approve or reject the application:
- Standard discount applications can be approved instantly.
- If the student requested a high discount (over 30%), the system flags it as "High Risk." The counselor can execute a **Manager Override** decision to authorize the seat.
- When the counselor clicks **Approve**, the database updates the status to "Approved."
- This trigger automatically syncs the contact details with the **Salesforce CRM Lead Pipeline** and sends a seat confirmation notice to the student's **WhatsApp via Twilio**.